Assignment 1—Supervised learning

A Self-Collected EEG-Audio Paired Dataset for Cross-Subject Supervised Auditory Envelope Reconstruction

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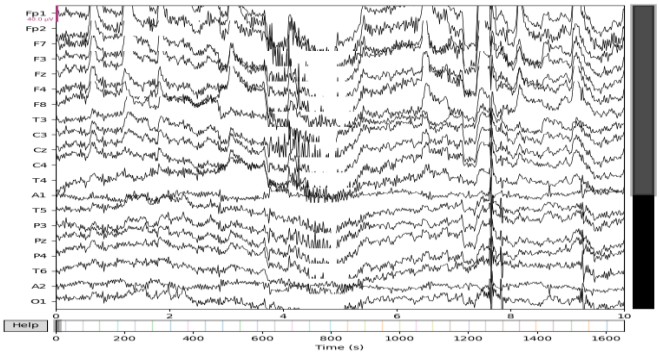
GitHub link：https://github.com/doctor-66/535505-AI-assignment1

Introduction

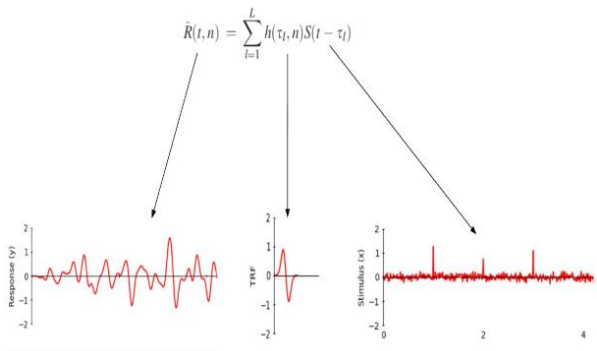
We construct a self-collected EEG-audio paired dataset to investigate whether EEG signals can reconstruct the corresponding speech envelope. This problem is relevant to auditory neural tracking, since cortical responses are known to follow temporal structures in continuous speech. At the same time, it is challenging because EEG signals are noisy, trial lengths are variable, and neural responses may differ substantially across subjects.



(Figure 1. Raw EEG data illustration)

The proposed dataset contains synchronized EEG recordings and speech-envelope targets collected from six subjects under two event contexts. In addition to paired neural and audio-derived targets, the dataset also preserves trigger-based temporal alignment, which makes it suitable for supervised regression and cross-subject evaluation. From a dataset perspective, its main value lies in the combination of self-collected multimodal signals, alignment between EEG and auditory targets, and the possibility of studying subject variability under a leave-one-subject-out setting.

Traditional temporal response function (TRF) models provide an interpretable linear mapping between continuous stimuli and neural responses (Figure 2). In this work, conventional TRF and deep TRF models are used mainly as feasibility demonstrations, in order to show that the proposed dataset can support both traditional supervised learning and more flexible deep-learning-based approaches.



(Figure 2. Temporal response function model)

Using leave-one-subject-out evaluation, this study addresses three questions:

1. Can EEG signals reconstruct the corresponding speech envelope?
2. Is cross-subject reconstruction possible on this dataset?
3. Can this dataset support both conventional TRF and deep-learning-based supervised methods?

We expect this dataset to serve as a useful benchmark for supervised EEG-audio regression, especially for studies of synchronized neural – acoustic representations, context-dependent analysis, and cross-subject generalization.

Research question and dataset motivation

This study is centered on a self-collected EEG-audio paired dataset and aims to evaluate whether such a dataset can support supervised learning for auditory neural tracking. Rather than focusing only on model performance, the main goal is to examine the value of the dataset itself, including its paired signal design, temporal alignment, variable-length structure, and cross-subject applicability.

Research question：

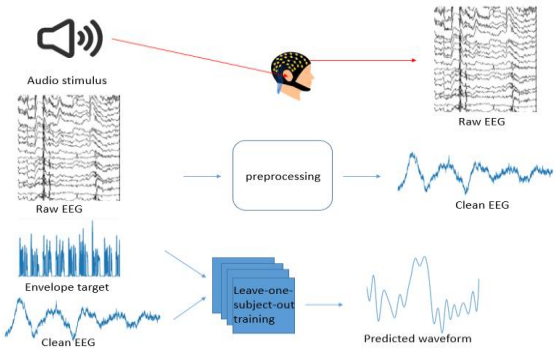
1. Can this self-collected EEG-audio dataset support supervised reconstruction of the corresponding speech envelope?
2. Can the dataset support cross-subject evaluation under leave-one-subject-out?
3. What dataset characteristics affect the feasibility of supervised learning on this task?

Dataset motivation：

This dataset is motivated by several characteristics. First, it provides paired neural signals and audio-derived envelope targets, which makes supervised regression feasible.

Second, the EEG recordings and audio targets are temporally aligned through trigger-based synchronization, allowing trial-level analysis of neural – acoustic relationships. Third, the trials are variable in length, which reflects a more realistic experimental condition and introduces an additional challenge for machine-learning pipelines. Fourth, the dataset contains two context conditions, making it possible to study context-dependent variation in neural responses. Finally, the dataset is designed for cross-subject learning, which makes it suitable for evaluating generalization under a leave-one-subject-out protocol.

Therefore, this dataset is not only a source of training samples, but also a structured benchmark for studying synchronized neural – acoustic representations, supervised EEG-to-envelope regression, and cross-subject generalization.

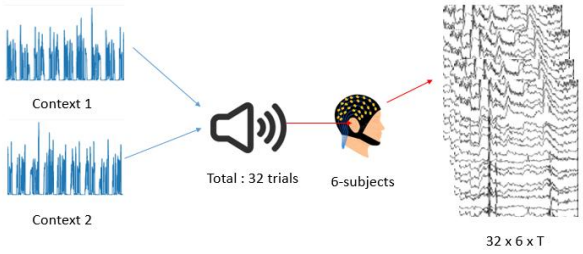


(Figure 3. Conceptual flow of implementation)

Dataset

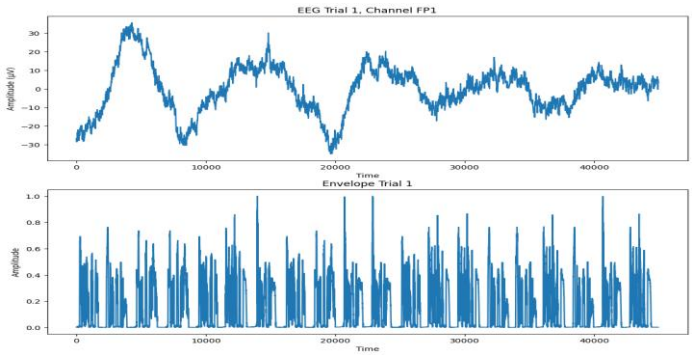
Data Overview：

The proposed dataset is a self-collected EEG-audio paired dataset designed for supervised learning on auditory neural tracking. It contains synchronized EEG recordings and speech-envelope targets collected from six subjects, with each subject contributing 32 trials. The dataset is non-public and was constructed specifically for studying whether neural signals can reconstruct corresponding auditory envelopes under a cross-subject setting.



(Figure 4. Overall structure of the self-collected EEG-audio paired dataset)

Each trial consists of multichannel EEG signals and an aligned one-dimensional speech envelope target. The original EEG recordings were acquired with 34 electrodes at a sampling rate of 1024 Hz, and after preprocessing, 30 channels were retained for analysis. The corresponding audio stimuli were spoken Mandarin number sequences, and their envelopes were extracted and temporally aligned to the EEG recordings. As a result, each sample in the dataset forms a paired structure of EEG input, envelope target, and context label.



(Figure 5. Paired sample of EEG and envelope)

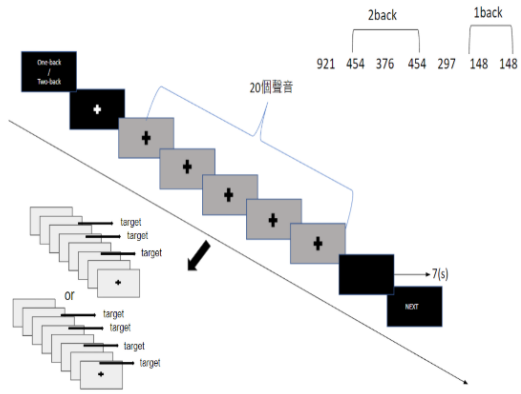
The dataset has several properties that make it suitable for supervised learning. First, it supports regression, since each EEG trial is paired with a continuous speech envelope target. Second, it includes two context conditions, enabling condition-aware analysis in addition to standard reconstruction. Third, the trial lengths are variable, which reflects realistic experimental recordings and introduces an additional challenge for data handling and model design. Finally, the dataset is naturally compatible with leave-one-subject-out evaluation, since data are organized at the subject level and each subject contributes multiple trials.

Overall, this dataset is intended not only as a source of labeled samples, but also as a structured benchmark for studying synchronized neural – acoustic representations, context-related variation, and cross-subject supervised learning in EEG-based auditory tasks.

Experimental paradigm and data collection

Experiment：

Subjects did an N-back-related cognitive experiment (Figure 6) while hearing spoken Mandarin number sequences. Each subject completed 32 trials, the stimuli consisted of sets of numbers pronounced in Mandarin, including morphemes such as “hundred” and “ten”. The trial lengths varied because the total audio duration differed across trials.



(Figure 6. N-back experiment diagram.)

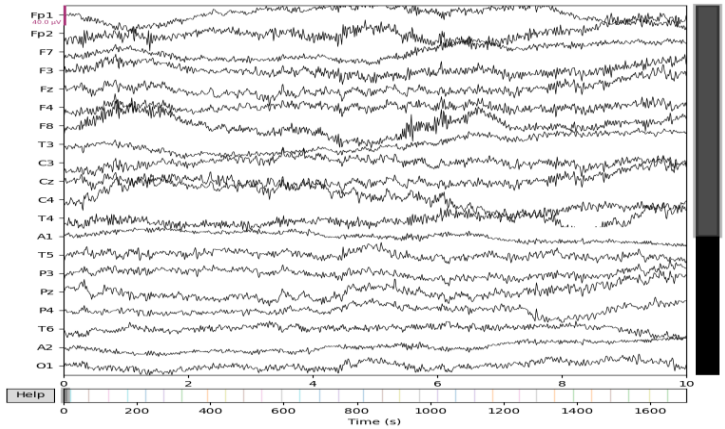
Recording environment and hardware:

- Quick-Cap
- 34 electrodes
- Grael 4K EEG system
- two computers for presentation and acquisition

Signal preprocessing, alignment, and target construction

EEG preprocessing：

EEG preprocessing：The EEG data were first processed by the band pass filter from 0.1 to 30 Hz. Then the Independent Component Analysis (ICA) was applied to remove the artifact of eye movement, muscle movement and other noise from the environment (Figure 7). Left and Right mastoid was selected as the offline re-reference channel, eye movement electrodes and mastoids were excluded from the data, the remain channel were 30. The process of the EEG was completed by the MNE-python tools.



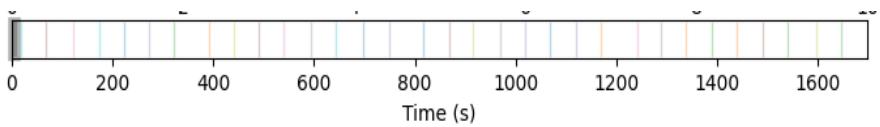
(Figure 7. Data after ICA.)

Audio processing and envelope extraction：

For extraction the envelope, the original sound file was a stereo type, and the original sampling rate was 44100 Hz, therefore the sound was first processed by low-pass filter with 512Hz to prevent aliasing. Re-sample to 1024 Hz as the EEG signal sampling rate. Hilbert transforms was applied to calculate the envelope of each trail sound, results of the envelope was only in one dimension and normalize between 0 and 1. The sound file was processed by scipy package.

Synchronization and labeling：

To alignment the EEG and envelope, during the recording of the EEG signal, the computer which used to present the experiment program is going to send a trigger to other computer which recording the signal will receive there is an event and marked in the timeline of signal in the file. Moreover, the trigger of the event had been marked down as the label (Figure 8).

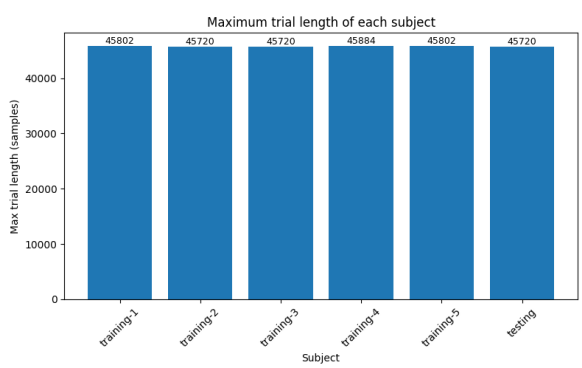


(Figure 8. Trigger in EEG data timeline.)

For the training method that using the hold trail, due to the length of each trail were different, the largest length in 32 trails selected and the others padding zero while having same length as the largest.

Dataset statistics and examples：

Per-subject max of trial lengths



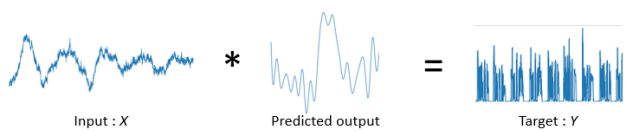
(Figure 9. Trial lengths across subjects.)

Context balance table

Subject	Context 1	Context 2	Total
training-1	16	16	32
training-2	16	16	32
training-3	16	16	32
training-4	16	16	32
training-5	16	16	32
testing	16	16	32

How the dataset can be used for supervised learning

The proposed dataset is suitable for supervised learning because each trial contains paired EEG signals and a corresponding speech-envelope target. In this work, the EEG recordings are treated as the input X, and the aligned speech envelope is treated as the regression target Y. Therefore, the dataset can directly support supervised EEG-to-envelope reconstruction (Figure 10).



(Figure 10. Expected reconstruction by model out and related input and target)

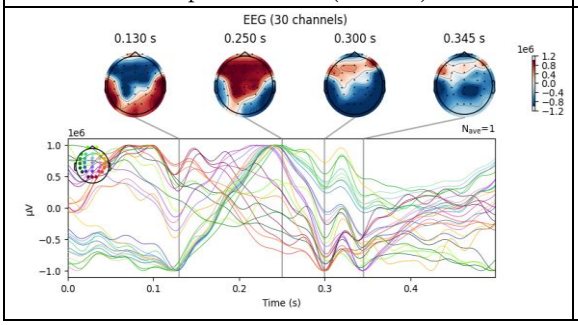
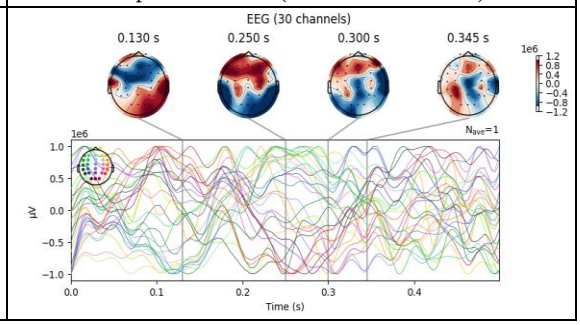
In addition, the dataset supports different data usage strategies. Since trials are variable in length, the data can be used either as full trials, which preserve complete temporal context, or as fixed-length windows, which simplify batch training and increase the number of samples. The two context conditions also make the dataset suitable for condition-aware analysis. Together with leave-one-subject-out evaluation, these properties make the dataset useful for supervised regression and cross-subject learning.

The loss function consists of four terms. The first term is the mean squared error (MSE), which measures the numerical difference between the predicted envelope and the ground-truth envelope. The second term is the correlation loss, defined as $1-\mathrm{Corr}(\hat{y},y)$, which encourages similarity in waveform shape and temporal trend. The third term is an L_2 regularization term on the residual TRF, which prevents the nonlinear residual component from becoming excessively large. The fourth term is a smoothness regularization term on the final TRF, which penalizes abrupt changes across adjacent time lags and improves interpretability. The total loss is defined as the weighted sum of these four terms.

$$\mathcal{L}_{total} = \mathcal{L}_{mse} + \alpha_{corr}\mathcal{L}_{corr} + \alpha_{res}\mathcal{L}_{res} + \alpha_{smooth}\mathcal{L}_{smooth}$$

Experimental Protocol and Compact Results

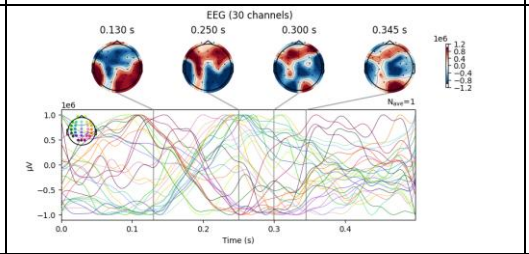
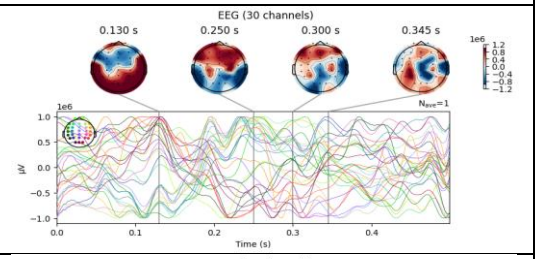
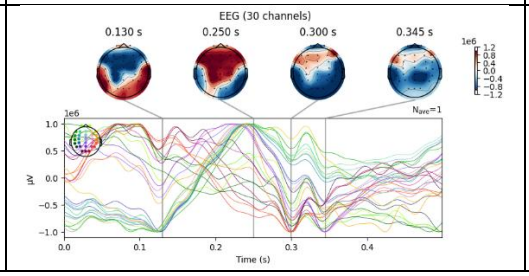
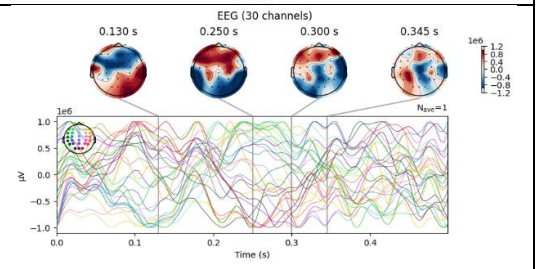
Model output：

Deep TRF model (full trial)	Deep TRF model (fixed-time window)
	

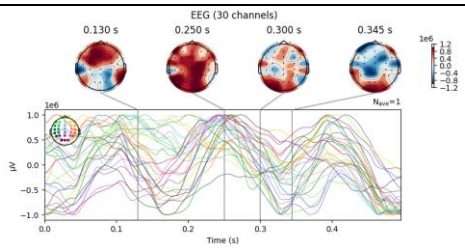
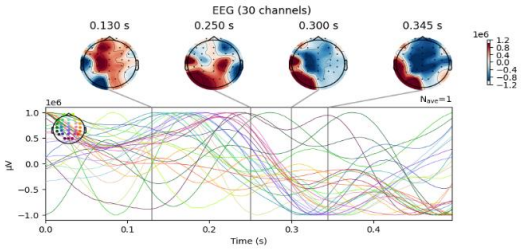
Quantitative result：

Method	Deep TRF model (full trial training)	Deep TRF model (fixed-time window)
Mean MSE (training)	0.054	0.055
Mean Corr (training)	0.023	0.099
Mean MSE (val)	0.054	0.055
Mean Corr (val)	0.006	0.050

Effect of training data amount：

Batch size	(full trial training)	(time window training)
32		
64		

Effect of with dimensionality reduction sampling rate (250 Hz)：

With dimensionality reduction (full trial training)	Without dimensionality reduction (time window training)
	

Dataset-Centered Analysis and Limitations

The model outputs suggest that the dataset preserves meaningful temporal structure related to auditory and task-associated cortical activity. Under both full-trial training and time-window training, the reconstructed responses showed plausible activity within expected temporal ranges, indicating that both data usage strategies are feasible. Full-trial training retains broader temporal context, while time-window training still captures locally informative neural patterns.

Changing the amount of training data through different batch sizes did not substantially alter the overall topographic structure. Response components associated with auditory evoked activity, P300, and working-memory-related activity remained visible across settings. The main effect of batch size was therefore numerical rather than structural: larger batch sizes led to more stable training, whereas smaller batch sizes produced greater fluctuation. Among the tested settings, batch size 64 with full-trial training gave the most stable result.

Subject-level analysis, however, revealed only limited differences across individuals. Although the model captured plausible temporal and spatial response patterns, these were largely shared across subjects, suggesting that the current dataset and training setup are more effective for learning common auditory response structure than fine-grained subject-specific variation. In addition, down-sampling as a dimensionality reduction strategy led to disorganized temporal and spatial patterns, suggesting that this approach removed information important for preserving interpretable neural responses.

Conclusion

This work presents a self-collected EEG-audio paired dataset for supervised auditory envelope reconstruction. Its main contribution lies in the dataset itself, including synchronized

EEG signals, aligned speech-envelope targets, two context conditions, variable-length trials, and subject-level organization suitable for leave-one-subject-out evaluation. These properties make the dataset a useful benchmark for studying supervised EEG-to-audio regression and cross-subject learning.

The current results suggest that the dataset preserves meaningful temporal structure related to auditory and task-associated cortical activity. Both full-trial training and time-window training produced plausible response patterns within expected temporal ranges, indicating that the dataset can support multiple supervised learning strategies. Batch size mainly affected numerical stability rather than overall temporal or topographic structure, with larger batch sizes giving more stable training.

However, subject-level analysis revealed only limited differences across individuals, suggesting that the current dataset and training setup are more effective for learning shared auditory response structure than fine-grained subject-specific variation. In addition, down-sampling led to disorganized temporal and spatial patterns, indicating that excessive dimensionality reduction may remove useful neural information.

Future work may improve the dataset in several practical directions. Increasing the number of subjects would strengthen cross-subject evaluation, while more diverse auditory stimuli would improve ecological validity. Since the current results favor richer temporal representations, future studies may explore moderate window overlap or milder feature compression instead of strong down-sampling. Subject-aware adaptation strategies may also help capture individual differences more effectively.

Overall, the dataset is not only feasible for supervised learning, but also useful for studying how data organization, temporal resolution, and subject variability influence EEG-based auditory reconstruction.

References

1. https://mne.tools/stable/index.html
2. https://eelbrain.readthedocs.io/en/stable/auto_examples/temporal-response-functions/trf_intro.html#sphx-glr-auto-examples-temporal-response-functions-trf-intro-py
3. Hjortkjær, J., Märcher-Rørsted, J., Fuglsang, S. A., & Dau, T. (2020). Cortical oscillations and entrainment in speech processing during working memory load.*European Journal of Neuroscience*,*51*(5), 1279-1289.

Appendix

Data Preprocessing

File：preprocess.ipynb

The preprocessing pipeline was designed to construct paired EEG – speech-envelope samples for supervised learning. Raw EEG signals were first loaded from the original recording files and filtered within the frequency range relevant to cortical auditory responses. The signals were then re-referenced, non-EEG channels were removed, and Independent Component Analysis (ICA) was applied to reduce structured artifacts such as eye movements. After artifact correction, only the EEG channels used for modeling were retained. In parallel, the corresponding speech waveform for each trial was filtered, resampled, and transformed into its amplitude envelope using the Hilbert transform, followed by normalization to reduce amplitude variation across trials. Finally, EEG segments were aligned with their matched speech envelopes on a trial-by-trial basis and stored together with the corresponding condition labels, forming the final paired dataset used for training and evaluation.

Model Training and Training Strategy

File：model_trf.py, train_trf_ft.py and train_trf_tw.py

The model used in this work was based on **TRFNet**, which combines a learnable global TRF with a nonlinear residual TRF branch. The EEG input was first passed through a shallow temporal encoder composed of 1D convolutional layers and dilated convolution blocks to extract temporal features. These features were then pooled and projected into a latent representation, from which a residual TRF was generated. The final TRF was obtained by adding this residual component to the global TRF through a learnable residual gate, allowing the model to preserve the interpretability of a classical TRF while improving flexibility through nonlinear modeling.

Two training strategies were implemented. The first was a **full-trial strategy**, in which each trial was kept in its original variable-length form. Because trial lengths differed across samples, the training code padded each batch and used a mask so that the loss was computed only on valid time points. The second was a **trial-window strategy**, in which each trial was segmented into fixed-length non-overlapping windows before training. This produced a larger number of uniform samples and simplified batch optimization. In both strategies, the reconstruction objective combined mean squared error, correlation-based loss, residual regularization, and TRF smoothness regularization. The best model was selected according to the lowest validation loss, and both the model checkpoint and training history were saved for later analysis.

Result Visualization

File：Plot.ipynb

To visualize the learned results, the trained model outputs were converted into forms suitable for interpretation at both the temporal and spatial levels. The learned global TRF and the difference between the final TRF and the global TRF were extracted and normalized for visualization. These TRF representations were then organized with standard EEG channel information and plotted using neurophysiological visualization tools, allowing the temporal dynamics and topographical distributions of the learned responses to be examined. In addition, the saved training history was loaded to plot the evolution of training and validation losses, providing a direct summary of convergence behavior and comparative performance under different training settings. Together, these visualizations helped interpret both the learned EEG-to-envelope mapping and the optimization behavior of the model.

Repository Structure

The repository is organized into four main parts: **data preprocessing**, **model definition**, **model training**, and **result storage/visualization**.

The preprocessing stage is mainly handled by `preprocess.ipynb`, which prepares the paired EEG and speech-envelope datasets from the raw recordings. This file performs signal cleaning, envelope extraction, and dataset construction. The accompanying file `ICA_note.txt` provides additional notes related to the ICA-based artifact removal procedure used during preprocessing. The processed outputs are saved as dataset files, including `training-1_dataset.zip` to `training-5_dataset.zip` and `testing_dataset.zip`, together with their corresponding `*_250.zip` versions for the 250-sample setting used in later experiments.

The core model architecture is defined in `model_trf.py`. This file implements the TRF-based neural network used throughout the project and serves as the shared model module for all training settings. Based on this model definition, two training scripts are provided. The file `train_trf_ft.py` is used for the **full-trial training strategy**, where variable-length trials are directly used for training with padding and masking. In contrast, `train_trf_tw.py` implements the **trial-window training strategy**, where each trial is divided into fixed-length windows before training. Both scripts load the processed datasets, train the model, perform validation, and save the best checkpoints together with the corresponding training histories.

The result folders, including `trf_2`, `trf_2_250`, and `trf_2_b32`, store the outputs of the full-trial training experiments under different settings. Similarly, `trf_tw_1`, `trf_tw_250`, and `trf_tw_b32` store the outputs of the trial-window experiments. These folders typically contain the trained model checkpoints and saved training histories, which are later used for analysis.

Finally, `plot.ipynb` is used for post-training visualization and interpretation. It reads the saved outputs from the training folders and generates figures such as TRF-related plots, topographical representations, and loss curves. In this way, the overall workflow of the repository follows a clear sequence: `preprocess.ipynb` prepares the data, `model_trf.py` defines the model, `train_trf_ft.py` and `train_trf_tw.py` perform training under different strategies, and `plot.ipynb` visualizes the final results.